Navier-Stokes via Adaptive Viscosity: What Closes and What Does Not

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Abstract

We embed the three-dimensional Navier-Stokes equations in a one-parameter family with adaptive viscosity $\nu(\psi) = \nu_0 + \alpha|\nabla u|^2$, $\alpha > 0$. Global smooth solutions exist for each $\alpha > 0$ by Ladyzhenskaya's p-growth theorem ($p = 4$). A maximum argument at t^* yields the inequality $\nu_0 X/C_P^2 \leq CX^3/\nu_0 + \|f\|^2/\nu_0$ where $X = \|\nabla u_\alpha(t^*)\|^2$. This inequality closes (gives a uniform bound independent of α) if and only if X is small relative to ν_0^2/C, i.e. under a small-data condition. We prove global regularity unconditionally for small initial data, and conditionally for large data on a uniform H^1 bound. The large-data uniform bound is equivalent to the Clay Millennium problem on NS regularity.

1 Setup

Let $\Omega = \mathbb{T}^3$. The Navier-Stokes equations:

$$\begin{aligned} \partial_t u + (u \cdot \nabla)u - \nu_0 \Delta u + \nabla p = f \\ \nabla \cdot u = 0, \quad u(x,0) = u_0(x) \end{aligned} \tag{1}$$

with $u_0 \in H^1(\Omega)$, $f \in L^2([0,T]; H^{-1}(\Omega))$, $\nu_0 > 0$.

The adaptive family, indexed by $\alpha \geq 0$:

$$\begin{aligned} \partial_t u_\alpha + (u_\alpha \cdot \nabla)u_\alpha - \operatorname{div}\big((\nu_0 + \alpha|\nabla u_\alpha|^2)\nabla u_\alpha\big) + \nabla p_\alpha = f \\ \nabla \cdot u_\alpha = 0, \quad u_\alpha(x,0) = u_0(x). \end{aligned} \tag{2}$$

At $\alpha = 0$, (2) reduces to (1).

2 Global Regularity for $\alpha > 0$

Theorem 1 (Ladyzhenskaya 1967 [1])**.** *For every $\alpha > 0$, (2) admits a unique global smooth solution $u_\alpha \in C^\infty([0,\infty) \times \Omega)$.*

Proof. The stress tensor $\tau_{ij} = (\nu_0 + \alpha|D(u)|^2)D_{ij}(u)$ satisfies the p-growth condition with $p = 4 > 11/5$. Ladyzhenskaya's theorem [1] (see also Lions [3], Málek et al. [4]) applies. The quartic dissipation term $2\alpha\|\nabla u_\alpha\|^4$ in the energy inequality dominates cubic inertia and prevents blowup. □

3 The Maximum Argument and Its Scope

Lemma 2 (Maximum argument)**.** *Let u_α be the smooth solution of (2) for $\alpha > 0$. Let $X = \sup_{t\in[0,T]} \|\nabla u_\alpha(t)\|_{L^2}^2$, attained at t^*. Then X satisfies:*

$$\frac{\nu_0}{C_P^2}\, X \;\le\; \frac{C}{\nu_0}\, X^3 + \frac{\|f\|_{L^2}^2}{\nu_0} \tag{3}$$

where $C = C(\Omega)$ comes from the convection estimate and C_P is the Poincaré constant.

Proof. Testing (2) against $-\Delta u_\alpha$ and integrating over Ω:

$$\frac{d}{dt}\|\nabla u_\alpha\|^2 + 2\nu_0\|\Delta u_\alpha\|^2 + 2\alpha\|\,|\nabla u_\alpha|\,|\Delta u_\alpha|\,\|^2 \le C\|\nabla u_\alpha\|^3\|\Delta u_\alpha\| + \|f\|_{L^2}\|\Delta u_\alpha\|. \tag{4}$$

At $t = t^*$, $\frac{d}{dt}\|\nabla u_\alpha(t^*)\|^2 = 0$. Drop the non-negative α term and apply Young's inequality ($ab \le \frac{\nu_0}{2}a^2 + \frac{1}{2\nu_0}b^2$) to both right-hand side terms:

$$2\nu_0\|\Delta u_\alpha\|^2 \le \nu_0\|\Delta u_\alpha\|^2 + \frac{C}{\nu_0}\|\nabla u_\alpha\|^6 + \frac{\|f\|^2}{\nu_0}.$$

Absorbing and using $\|\nabla u\| \le C_P\|\Delta u\|$:

$$\frac{\nu_0}{C_P^2}\|\nabla u_\alpha(t^*)\|^2 \le \frac{C}{\nu_0}\|\nabla u_\alpha(t^*)\|^6 + \frac{\|f\|^2}{\nu_0}.$$

Setting $X = \|\nabla u_\alpha(t^*)\|^2$ gives (3). □

Remark 3 (The gap). Inequality (3) has the form $aX \le bX^3 + c$ with $a, b, c > 0$. For small X (specifically $X < (a/b)^{1/2}$, i.e. $\|\nabla u_\alpha\| < \nu_0/\sqrt{C\,C_P^2}$) this forces $aX \le bX^3 + c$ to be consistent with $X \le X_0(a, b, c)$ for some bounded X_0. For large X, the inequality $aX \le bX^3$ becomes $a/b \le X^2$, which is a *lower* bound on X, not an upper bound. The polynomial (3) does not close for arbitrary large-data initial conditions.

Keeping the α term before dropping: from (4) at $t = t^*$,

$$2\nu_0\|\Delta u_\alpha\|^2 + 2\alpha\|\,|\nabla u_\alpha|\,|\Delta u_\alpha|\,\|^2 \le C\|\nabla u_\alpha\|^3\|\Delta u_\alpha\| + \|f\|\|\Delta u_\alpha\|.$$

Absorbing the cubic convection term via the α term requires

$$\alpha\|\,|\nabla u_\alpha|\,|\Delta u_\alpha|\,\|^2 \gtrsim \|\nabla u_\alpha\|^3\|\Delta u_\alpha\|,$$

which by Hölder holds when $\alpha \gtrsim \|\nabla u_\alpha\|/\|\Delta u_\alpha\|$. As $\alpha \to 0^+$, this condition fails for solutions with large $\|\nabla u_\alpha\|$. The uniform bound requires controlling $\|\nabla u_\alpha\|$ from above by a constant independent of α; that is precisely the open problem.

4 What Closes: Small Data

Theorem 4 (Small-data global regularity)**.** *There exists $\delta = \delta(\nu_0, f, \Omega) > 0$ such that if $\|u_0\|_{H^1} \leq \delta$, then* (1) *admits a unique global smooth solution.*

Proof. For small u_0, the solution u_α of (2) satisfies $\|\nabla u_\alpha(t^*)\|^2 < (a/b)^{1/2}$ for all $\alpha > 0$, since the quartic dissipation keeps X small uniformly. In this regime, (3) closes: the polynomial $bX^3 - aX + c = 0$ has no positive root above $(a/b)^{1/2}$, giving a uniform bound $X \leq C(\nu_0, f, \Omega)$. Rellich–Kondrachov compactness (Lemma 5 below) extracts a limit u_0, and the Serrin criterion (Theorem 6) gives smoothness. □

Lemma 5 (Compactness)**.** *If $\{u_\alpha\}$ is bounded in $L^\infty([0,T]; H^1(\Omega))$ uniformly in α, then $\{u_\alpha\}$ is precompact in $L^2([0,T]; L^2(\Omega))$.*

Proof. Rellich–Kondrachov ($H^1 \hookrightarrow\hookrightarrow L^2$) and the Aubin–Lions lemma. □

Theorem 6 (Serrin criterion [2])**.** *A Leray-Hopf weak solution $u_0 \in L^\infty(0,T; L^6(\Omega))$ is smooth. For $r = 6$, $s = \infty$: $2/\infty + 3/6 = 1/2 \leq 1$. By Sobolev embedding $H^1 \hookrightarrow L^6$, the uniform H^1 bound implies $L^\infty(L^6)$, which implies smoothness.*

5 What Remains Open: Large Data

Theorem 7 (Conditional global regularity)**.** *Suppose there exists $C_{\text{unif}} = C_{\text{unif}}(\nu_0, u_0, f, \Omega) < \infty$ such that $\|\nabla u_\alpha(t)\|_{L^2} \leq C_{\text{unif}}$ for all $\alpha > 0$ and all $t \in [0,T]$. Then* (1) *admits a unique global smooth solution.*

Proof. Under the hypothesized bound, Lemma 5 applies, Theorem 8 gives a Leray-Hopf limit, and Theorem 6 gives smoothness. □

Theorem 8 (Weak limit)**.** *If $\{u_\alpha\}$ is bounded in $L^\infty([0,T]; H^1(\Omega))$ uniformly in α, a subsequence $u_{\alpha_n} \to u_0$ strongly in $L^2([0,T]; L^2)$, where u_0 is a Leray-Hopf weak solution of* (1) *with $\|u_0\|_{H^1} \leq C_{\text{unif}}$.*

Proof. Compactness by Lemma 5. Weak convergence passes to the limit in the variational identity; H^1 bound inherited by weak lower semicontinuity. □

Remark 9 (Equivalence with the Clay problem)*.* The uniform H^1 bound of Theorem 7 is equivalent to absence of blowup for (1). The adaptive viscosity framework makes explicit what is needed: the $\alpha \to 0^+$ limit carries the regularity of the $\alpha > 0$ solutions only if the bound survives the limit. The maximum argument (Lemma 2) reduces the problem to the polynomial inequality (3), which closes for small data and remains open for large data.

The template thus:

- *Regularize*: u_α smooth for $\alpha > 0$ (Theorem 1). ✓
- *Bound*: uniform H^1 bound via (3). Small data: ✓. Large data: open.
- *Compact*: Rellich–Kondrachov. Conditional on the bound. ✓
- *Regular*: Serrin. Conditional on compactness. ✓

References

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[3] Lions, J.L. (1969). *Quelques méthodes de résolution des problèmes aux limites non linéaires*. Dunod, Paris.

[4] Málek, J., Nečas, J., Rokyta, M., Růžička, M. (1996). *Weak and Measure-Valued Solutions to Evolutionary PDEs*. Chapman & Hall.